HDOC DAY 46

PROBLEM

Write a C program that reads the *size* and elements of a variable-length integer array, *nums[]*. The array may contain duplicate elements. Pass the array to a function named *Bubble_sort()* using its address and sort the elements in ascending order using the Bubble Sort technique. After sorting, pass the sorted array to another function named *occurrence()*. Use pointers to pass the array between functions. The function *occurrence()* must:

1. Read an integer *target*.
2. Find the first and last occurrences of *target* in the sorted array.
3. Display the values and indices of the first and last occurrences.
4. Display -1, -1 if the *target* is not present.

CODE

```
#include <stdio.h>


void Bubble_sort(int *nums, int n)
{
   int i, j, temp;


   for (i = 0; i < n - 1; i++)
   {
      for (j = 0; j < n - i - 1; j++)
      {
         if (*(nums + j) > *(nums + j + 1))
         {
            temp = *(nums + j);
```

```
            *(nums + j) = *(nums + j + 1);
            *(nums + j + 1) = temp;
        }
    }
  }
}

void occurrence(int *nums, int n)
{
  int target, first = -1, last = -1, i;

  printf("Enter target: ");
  scanf("%d", &target);

  for (i = 0; i < n; i++)
  {
    if (*(nums + i) == target)
    {
      if (first == -1)
        first = i;

      last = i;
    }
  }

  if (first == -1)
  {
    printf("-1, -1\n");
```

```
    }
    else
    {
        printf("First occurrence: value = %d, index = %d\n",
            target, first);
        printf("Last occurrence: value = %d, index = %d\n",
            target, last);
    }
}

int main()
{
    int n, i;

    printf("Enter size of array: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Invalid array size.\n");
        return 1;
    }

    int nums[n];

    printf("Enter array elements: ");
    for (i = 0; i < n; i++)
    {
```

```
        scanf("%d", nums + i);
    }

    Bubble_sort(nums, n);

    printf("Sorted array: ");
    for (i = 0; i < n; i++)
    {
        printf("%d ", *(nums + i));
    }
    printf("\n");

    occurrence(nums, n);

    return 0;
}
```